

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. MTB-502: Abstract Algebra

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) If G is the Klein four-group, then show that $\text{Aut}(G) \cong S_3$. [3]
- (b) Define characteristic of a ring. Does there exist a ring with characteristic n , for every $n \geq 0$? [2]
- (c) If $f : R \rightarrow R'$ is an onto ring homomorphism, then show that if R is commutative, so is R' . [2]
- (d) What is the quotient field of a finite integral domain? [2]
- (e) Define commutator of a group G . Show that for $G = S_3$, the commutator subgroup G' is A_3 . [3]
- (f) Define group action on a set. Let H be a normal subgroup of G and $A = G/H$. Define $*$: $G \times A \rightarrow A$ by $g * a = gag^{-1}$ for $g \in G, a \in A$. Show that $*$ is a group action. [2]

Question 2.

- (a) Let $\text{Aut}(G)$ be the set of all automorphisms of G and $\text{Inn}(G)$ be the set of all inner automorphisms of G . Show that $\text{Aut}(G)$ is a group with respect to composition of mappings and $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$. [8]
- (b) Let G be an infinite cyclic group. Find the order of $\text{Aut}(G)$. [6]

Question 3.

- (a) Let G be any group and S be any set. Prove that any group homomorphism from G to $A(S)$, the symmetric group of S , defines an action of G on S , and conversely every action of G on S induces a homomorphism $G \rightarrow A(S)$. Deduce from this Cayley's theorem. [8]
- (b) (i) Show that the order of the group of automorphisms of a finite group of order n is a divisor of $n!$. [3]
(ii) Let G be a group. Show that the commutator subgroup G' of G is the smallest subgroup of G such that G/G' is abelian. [3]

Question 4.

- (a) Define a unit in a ring. Show that in a ring R with unity 1, the units form a group under multiplication. Find the group of units in $\mathbb{Z}_8$. Is this group isomorphic to the Klein four-group? [1+4+3+2]
- (b) Prove that a commutative ring R is without zero divisors if and only if for all $a, b, c \in R$, if $ab = ac$ and $a \neq 0$, then $b = c$. [4]

Question 5.

- (a) Define characteristic of a ring. Show that the characteristic of an integral domain is either prime or zero. [1+5]
- (b) Define prime ideal and maximal ideal of a ring. Find all prime ideals and maximal ideals in the ring $\mathbb{Z}_6$. [1+1+6]

Question 6.

- (a) Let R be a commutative ring with unity. Show that an ideal M of R is a maximal ideal of R if and only if the quotient ring R/M is a field. [6]
- (b) Define the quotient field of an integral domain. Give an example to show that non-isomorphic integral domains may have isomorphic fields of quotients. [1+2]
- (c) State and prove the Fundamental Theorem of Ring Homomorphism. [5]

Question 7.

- (a) Let $f(x)$ be any polynomial and $g(x)$ be a non-zero polynomial in the polynomial domain $F[x]$ over a field F . Show that there exist unique polynomials $q(x)$ and $r(x)$ in $F[x]$ such that

$$f(x) = g(x)q(x) + r(x)$$

where either $r(x) = 0$ or $\deg r(x) < \deg g(x)$. [8]

- (b) Show that every Euclidean domain is a principal ideal domain. [6]